



Spendly

Offline-First · Zero- Knowledge · Peer-to- Peer Finance Ecosystem

A semester VII major project in Information Technology — redefining how financial data is stored, shared, and secured. No cloud. No exposure. No compromise.

[Student Name] · [Enrollment Number] · Semester VII · Information Technology ·
[College Name]

THE PROBLEM

Finance Is Broken by Design

Every transaction you make today is stored on someone else's server, exposed to breaches, and dependent on a connection you may not have. The infrastructure is fragile — and your data pays the price.

Cloud Dependency

Every transaction requires internet. No connectivity means no commerce.

Privacy Exposure

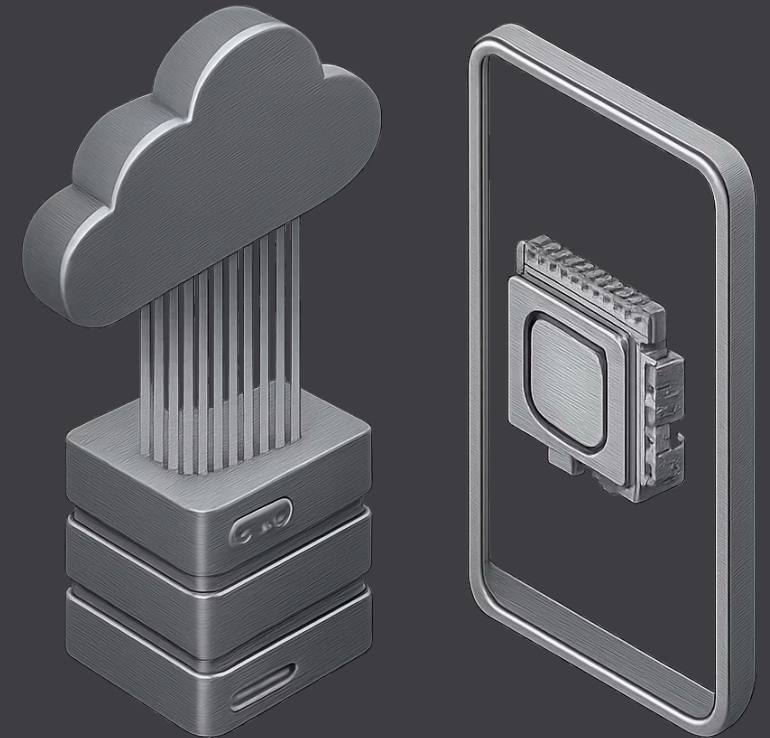
Centralized databases are single points of failure. Breaches are inevitable.

Recurring Costs

Monthly POS subscriptions and server fees burden small merchants.

Data Breaches

Financial records on third-party servers are permanently at risk.



THE SOLUTION

Three Apps. One Ecosystem. Zero Compromise.

Spendly is not a single application — it is a coordinated ecosystem of three purpose-built tools, each designed to operate entirely offline, communicate peer-to-peer, and never expose your data to a central authority.

Spendly

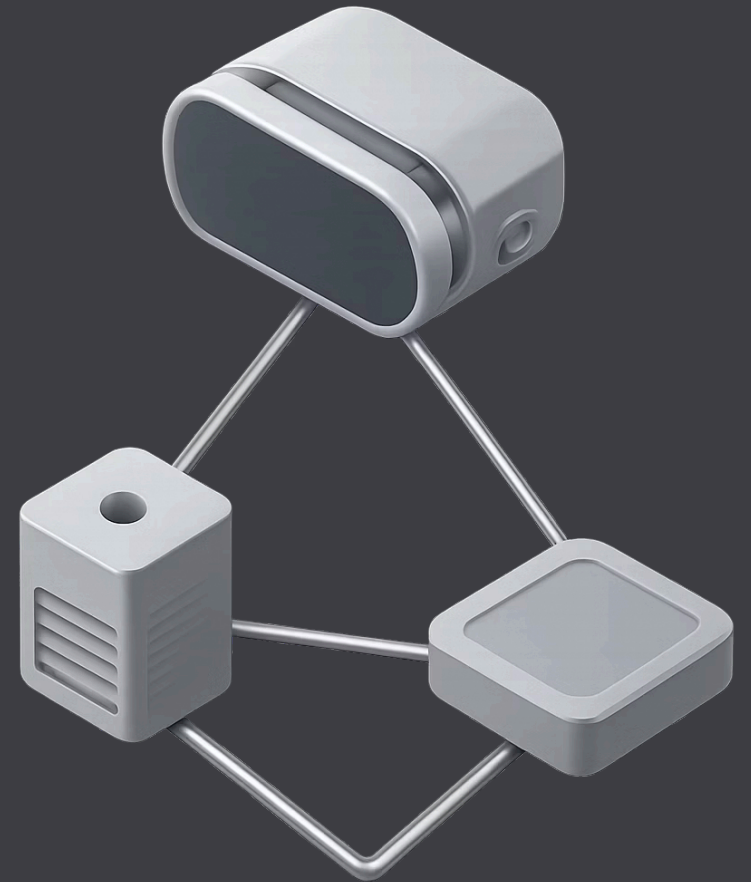
The consumer wallet. Track expenses, claim coupons, generate invoices, and manage your financial life — all locally, all privately.

Spendly Shop

The merchant terminal. Accept payments, manage inventory, issue QR invoices, and sync offline — with zero monthly fees.

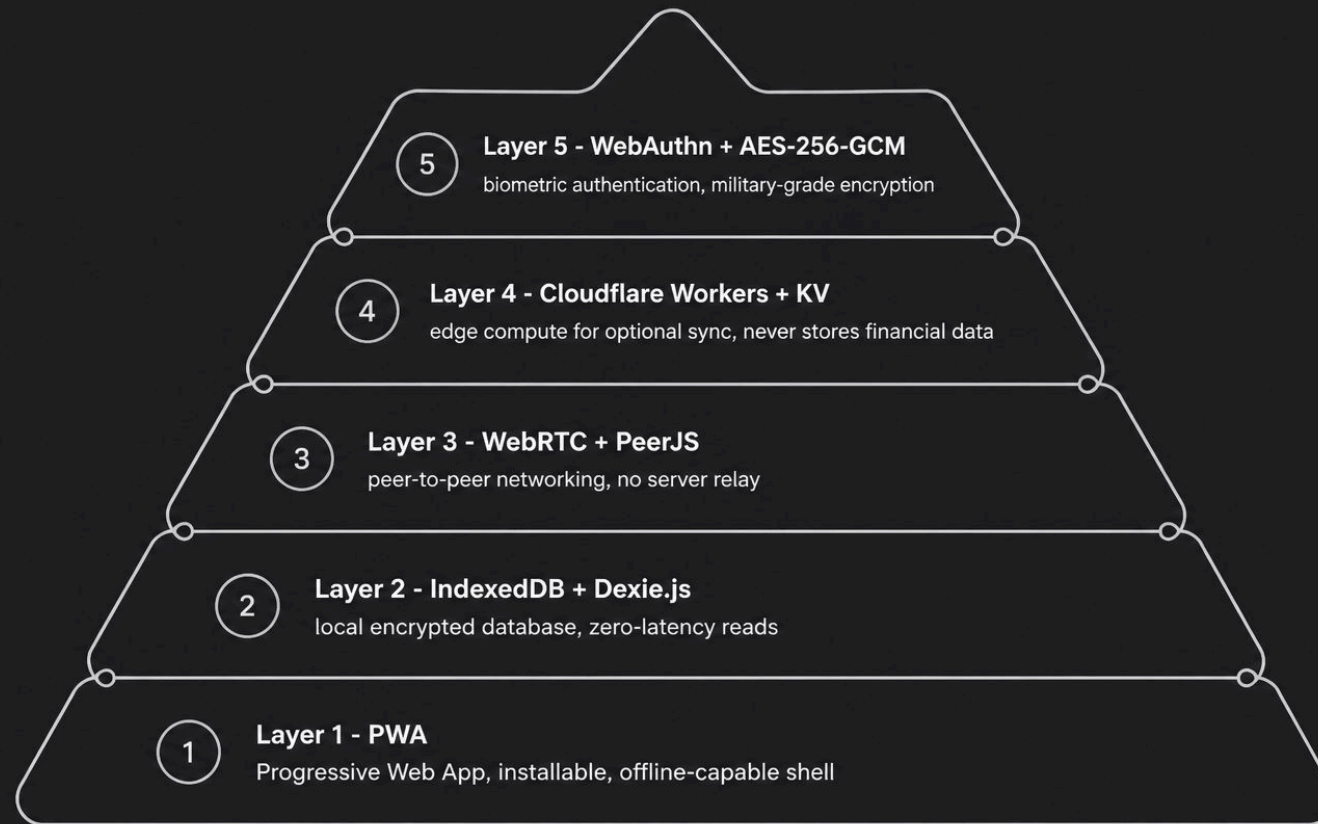
Spendly Radar

The discovery layer. Find nearby offers, validate coupons, and connect to merchants through encrypted peer-to-peer signals.



Layered Like Silicon. Secured Like a Vault.

Spendly's architecture is modeled after the precision of Apple's silicon stack — each layer has a singular responsibility, communicates through clean interfaces, and operates independently. The result is a system that is fast, resilient, and impossible to compromise from the outside.



Each layer is independently replaceable. The system degrades gracefully — even with all external services offline, core financial functions remain fully operational.

USER FLOW

From Discovery to Analytics — In Seconds.

Spendly's transaction flow eliminates every unnecessary step. A customer discovers an offer, claims it, scans a barcode, and receives a validated invoice — all without internet, all without exposing personal data to any server.

Offer Discovery

Radar detects nearby merchant offers via peer-to-peer signal.

Coupon Claim

Customer claims offer locally — no server round-trip required.

Barcode Scan

Product lookup and price validation happen on-device.

Invoice Generation

Encrypted invoice created and shared peer-to-peer instantly.

Expense Analytics

Real-time savings and spending insights, computed locally.

TECHNOLOGY

Every Technology Chosen With Intent.

Spendly's stack is not a collection of popular tools — it is a deliberate assembly of technologies that share a single philosophy: operate at the edge, respect privacy, and never depend on a central authority. Each component was selected for its offline-first capabilities, security guarantees, or peer-to-peer architecture.

Frontend

- React — component-driven UI
- PWA — installable, offline shell
- WebAuthn — passwordless biometric auth

Core Stack

- **Database:** IndexedDB + Dexie.js — encrypted local storage
- **Networking:** WebRTC + PeerJS — direct device-to-device
- **Edge:** Cloudflare Workers + KV — optional sync layer
- **Security:** AES-256-GCM — authenticated encryption

SECURITY

Your Data Never Leaves Your Device.

Spendly implements a zero-knowledge architecture by design. The server knows nothing about your transactions, your balance, or your identity. Every piece of financial data is encrypted with AES-256-GCM before it is written to storage — and the decryption key never leaves your secure enclave.

Zero-Knowledge

No server ever sees your financial data. Not in transit. Not at rest.

Offline-First

All operations execute locally. Connectivity is optional, not required.

WebAuthn

Biometric authentication replaces passwords. Your face is your key.

AES-256-GCM

Military-grade authenticated encryption for every stored record.



Made with **GAMMA**

FEATURES

Everything a Modern Finance Ecosystem Demands.

Spendly delivers a complete financial toolkit — for consumers and merchants alike. Every feature operates without internet, without a subscription, and without exposing your data to third parties. This is what finance looks like when it is built for the real world.



Expense Tracking

Real-time categorization and spending insights, computed entirely on-device.



Merchant Billing

Generate, sign, and share encrypted invoices without any POS hardware.



Cash Wallet + EMI

Track cash balances, set EMI alerts, and manage payment schedules offline.



QR Invoice + Offline Sync

Share invoices via QR. Sync peer-to-peer when connectivity is available.



Built for Today. Designed for Tomorrow.

Core Advantages

- **Offline-First** — fully functional without internet
- **Zero Server Cost** — no monthly POS or cloud fees
- **Privacy by Default** — zero-knowledge architecture
- **Fast Billing** — local computation, no latency
- **P2P Communication** — direct, encrypted device links

Future Scope

- **AI Insights** — on-device spending intelligence
- **Voice Billing** — hands-free merchant transactions
- **UPI + NFC** — native payment rail integration
- **GST Compliance** — automated tax invoice generation
- **Multi-Store** — unified merchant dashboard

Spendly.

Privacy. Innovation. Offline-First Finance.

Spendly is more than a capstone project — it is a statement about what financial technology should be. Local-first. Zero-knowledge. Peer-to-peer. Built for a world where connectivity is unreliable, privacy is non-negotiable, and small merchants deserve enterprise-grade tools without enterprise-grade costs.

Thank You

Questions are welcome. The Spendly ecosystem is open for discussion.

